

Brain Tumor MRI Classification (Images)

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4-Class (Multiclass classification problem) · Format 2

Train and compare classification models

SKImage + OpenCV → Feature Extraction

01 K-Nearest Neighbors

02 Support Vector Machine

03 Random Forest

Raw Pixels → Learns Features Automatically

04 Custom CNN + Grad-CAM

*Conv layers auto-extract spatial features
Grad-CAM heatmap shows model attention*

05 ResNet-50 Transfer + LIME

*Pretrained ImageNet weights transferred
LIME explains patch-level predictions*